

Noise Robust Classification of Moving Vehicles via Micro-Doppler Signatures

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Abstract—For robust classification of moving wheeled and tracked vehicles using returned micro-Doppler signals within short dwell time, the influence of receiver white noise and low spectrum resolution are encountered. In this paper, noise reduction and super-resolution are realized simultaneously via a redundant dictionary based l_1 -norm optimization method. Experiments based on the measured data are presented, including the analysis of noise reduction performance, and the evaluation of classification robustness for different signal-to-noise ratio cases. The experimental results are also compared with related methods.

I. INTRODUCTION

The wheeled and tracked vehicles often undertake different tasks in battlefield, thus the discrimination of the two types of vehicles is important to a successful tactic. Considering the realization of multifunction for modern ground surveillance radar in detection and classification of moving vehicles, the research on classification within short dwell time is a promising way to balance the contradiction of dwell time between the search and classification functions. Since the typical moving structure of vehicle can be modeled by a special Doppler modulation called micro-Doppler effect[1], the micro-Doppler information contained in the radar returned signals can be utilized to discriminate different moving vehicles [2], [3].

For moving vehicle classification within short dwell time, the classification performance usually descends with the reduction of signal-to-noise ratio (SNR), and dramatically deteriorates especially in low SNR case [4]. Since the SNR directly relates to the distance between the target and radar for a given noise power and radar power, noise robustness of a classification method is an important factor in increasing the classification distance between the target and radar in the real application. However, to the best of our knowledge, noise robustness is not considered in the existing work for the moving vehicle classification. In addition, due to the short dwell time, the resolution of Doppler spectrum is low. As a result, a dominant bulk motion component will have wide main lobe and sidelobes, which will affect nearby micro-

Doppler components. The research reported here seeks an optimization method to deal with the above-mentioned questions. The rest content of this paper is organized as follows. We analyze the micro-Doppler signatures of moving vehicles in Section II. The sparse representation approach, which realizes the super-resolution spectrum and reduces the receiver white noise in the returned echo, is introduced in Section III. The performance evaluation, classification results and corresponding analyses based on the measured data are given in Section IV, followed by the conclusion in Section V.

II. ANALYSIS OF MICRO-DOPPLER SIGNATURES FOR WHEELED AND TRACKED VEHICLES

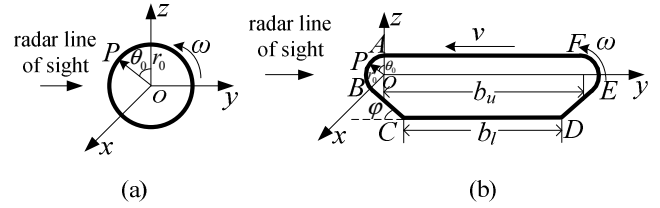


Figure 1. Micro motion structures of the wheeled and tracked vehicles. (a) A moving wheel. (b) A moving track.

In the following analysis, a local coordinate which has the same translation motion as the vehicle mainbody is introduced to depict the micro motion of wheel or track as shown in Figure 1. In such coordinate, the radial bulk velocity of vehicle, namely v , has been compensated. For a moving wheel as shown in Figure 1a, assume that K scattering points distribute evenly on the wheeled edge, the returned micro motion signal of the wheel is

$$s(t) = \sum_{k=1}^K \xi_k \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn(\omega t + \theta_k)} \quad (1)$$

where $\beta = 4\pi r_0 / \lambda$ and $\theta_k = \theta_0 + 2\pi(k-1)/K$, J_n denotes the n th-order Bessel function of the first kind, ω , r_0 , λ and t denote the angle frequency of rotation, the radius of rotation,

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wavelength and time respectively, θ_0 is the initial angle of reference scattering point P , ξ_k and θ_k are the scattering coefficient and the initial angle of the k th scattering point respectively [5].

In the similar way, the returned micro motion signal of the moving track as shown in Figure 1b is equal to the sum of returned signals generated by parts AB , BC , CD , DE , EF and FA , which is represented as

$$\begin{aligned}
 s(t) &= s_{AB}(t) + s_{BC}(t) + s_{CD}(t) + s_{DE}(t) + s_{EF}(t) + s_{FA}(t) \\
 &= \sum_{k=1}^{K_i} \xi_k \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn(\omega t + \theta_k^i)} \\
 &\quad + \sum_{k=1}^{K_{ii}} \xi_k \exp[-j \frac{4\pi}{\lambda} (y_k^{ii} + vt \cos \varphi)] \\
 &\quad + \sum_{k=1}^{K_{iii}} \xi_k \exp[-j \frac{4\pi}{\lambda} (y_k^{iii} + vt)] \\
 &\quad + \sum_{k=1}^{K_{iv}} \xi_k \exp[-j \frac{4\pi}{\lambda} (y_k^{iv} + vt \cos \varphi)] \\
 &\quad + e^{j \frac{4\pi b_u}{\lambda}} \sum_{k=1}^{K_v} \xi_k \sum_{n=-\infty}^{\infty} J_n(\beta) e^{jn(\omega t + \theta_k^v)} \\
 &\quad + \sum_{k=1}^{K_{vi}} \xi_k \exp[-j \frac{4\pi}{\lambda} (y_k^{vi} - vt)]
 \end{aligned} \tag{2}$$

where b_u is the length of the upper track, φ is the oblique angle of the track, $K_i, K_{ii}, K_{iii}, K_{iv}, K_v$ and K_{vi} denote the scattering point numbers of parts AB , BC , CD , DE , EF and FA , respectively. θ_k^u with $u = i, v$ denotes the initial angle of the k th evenly distributed scattering point from AB or EF respectively and y_k^u with $u = ii, iii, iv, vi$ denotes the initial location of the k th evenly distributed scattering point from BC , CD , DE or FA , respectively.

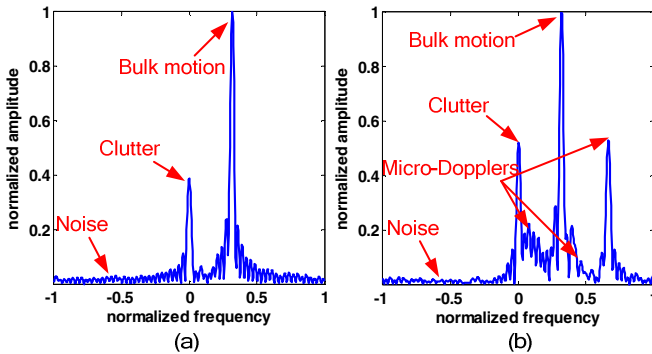


Figure 2. Doppler spectra of moving vehicles from the measured data. (a) Wheeled vehicle. (b) Tracked vehicle.

When bulk motion is considered, the phase term $e^{j \frac{4\pi}{\lambda} vt}$ induced by the bulk velocity should be introduced in the returned micro-Doppler echo as in (1) or (2). The micro-Doppler frequencies will hence distribute around the Doppler

frequency induced by the bulk velocity. Two Doppler spectrum samples from measured data are shown in Figure 2. Besides micro-Doppler frequencies, there are three kinds of main frequency components in Doppler domain for moving vehicles, i.e. the bulk motion component, the ground clutter and the receiver white noise. As shown in Fig 2a, the micro-Doppler frequencies of the wheeled vehicle are barely observed due to the rubbery material of the wheel. Contrast with the wheeled vehicle, the tracked vehicle has remarkable micro-Doppler frequencies induced by the track as shown in Fig 2b. The micro-Doppler components distribute in the frequency band corresponding to velocity range $[0, 2v]$.

III. SPARSE REPRESENTATION OF MICRO-DOPPLER SIGNATURES

Since the returned radar signals often contain ground clutter, the clutter should be suppressed properly for the next classification stage. In this paper, a harmonic extraction technique known as CLEAN is used to suppress the ground clutter distributing around 0 frequency[6]. Given a micro-Doppler signal $\mathbf{x}$, the clutter suppressed can be realized as follows.

- 1) Set the bandwidth of clutter in Doppler domain and the maximum iteration number. Use micro-Doppler signal $\mathbf{x}$ as input signal $\mathbf{x}_m$.
- 2) Calculate the Doppler spectrum of $\mathbf{x}_m$ and find its maxima point in the bandwidth of clutter.
- 3) Calculate the corresponding Doppler frequency f_d based on the position of the maxima point, and record the corresponding amplitude ρ and phase ϕ .
- 4) Reconstruct the largest harmonic wave of clutter as $\mathbf{x}_f(t) = (\rho/T) \exp[j(2\pi f_d t + \phi)]$.
- 5) Subtract the contribution of the detected frequency component from the input signal: $\mathbf{x}_{new} = \mathbf{x}_m - \mathbf{x}_f$.
- 6) Use $\mathbf{x}_{new}$ as new input signal, and repeat step 2 to step 5 until the maximum iteration number is met.

After the clutter suppression, the main components of the Doppler spectrum are white noise and target echo. Thus noise reduction should be considered in the next process. In addition, since the short dwell time, the main lobe and sidelobes of the dominant bulk motion component are both wide. As a result, the nearby micro-Doppler components will be affected, and the information contained in these micro-Doppler components is hence lost. In order to take full advantage of the micro-Doppler information, super-resolution methods should be utilized. Usually, the micro-Doppler signature of the moving vehicle is band limited, the Doppler frequencies of the moving vehicle only occupy a fraction of the entire Doppler frequency domain, which corresponds to the velocity range $[0, 2v]$. This indicates the sparsity of the micro-Doppler signature in Doppler frequency domain, and its sparse representation can be achieved via optimization method used in sparse reconstruction or compressed sensing (CS) [7], [8]. From (1) and (2), the returned signals of the wheeled and tracked parts can be represented by Fourier series. Thus the sparse representation can be obtained by solving a l_1 -norm constrained optimization problem based on Fourier basis, i.e.,

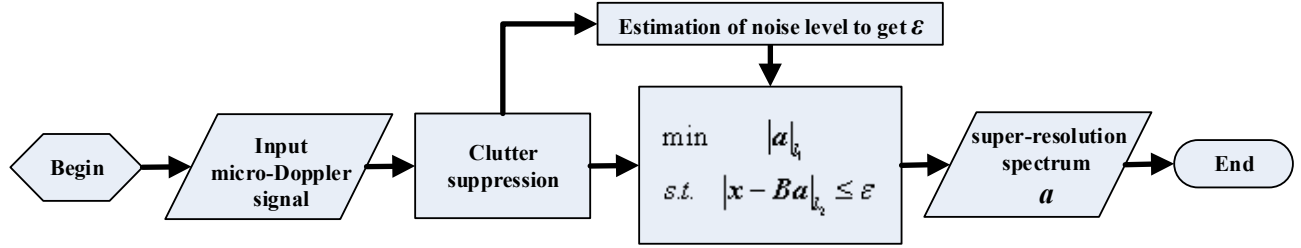


Figure 3. Flow chart of sparse representation of a micro-Doppler signal.

given a micro-Doppler signal $\mathbf{x}$, its sparse representation, namely $\mathbf{a}$, is obtained as

$$\min \|\mathbf{a}\|_{l_1}, \quad \text{s.t.} \quad \|\mathbf{x} - \mathbf{B}\mathbf{a}\|_{l_2} \leq \varepsilon \quad (3)$$

where $\|\cdot\|_{l_1}$ and $\|\cdot\|_{l_2}$ denotes the l_1 -norm and l_2 -norm of a vector respectively, $\mathbf{B}$ is the Fourier basis matrix, ε is reconstruction error. Orthogonal Matching Pursuit (OMP) [7] is employed in this paper as the sparse recovery techniques for solving the optimization problem in (3).

In order to realize super-resolution and noise reduction via (3), the selection of $\mathbf{B}$ and ε should be considered. The general entry of the $N \times M$ basis matrix $\mathbf{B}$ in (3) is

$$B(n, m) = \exp(j 2\pi n m / N) \quad (4)$$

where $n=0,1,\dots,N-1$, N is the length of the signal $\mathbf{x}$ and $m=0,1,\dots,M-1$, M is the length of the Doppler spectrum. When $N < M$, the sparse representation could obtain the super-resolution ability due to the redundancy of $\mathbf{B}$.

As for ε , since the reconstruction error in (3) allows the l_1 -norm term to impose some small frequency components to approximation to 0, noise reduction can be achieved provided that the reconstruction error is associated with the noise level. In our work, since the frequency bandwidth of moving vehicle corresponds to velocity range $[0, 2v]$ in the Doppler frequency domain, the rest frequency bins can be considered as noise range which can be used to estimate ε for a specific signal.

Accordingly, a flow chart of the proposed super-resolution combined with clutter suppression and noise reduction is given in Figure 3.

IV. EXPERIMENT RESULTS

In this experiment, we utilize measured data to validate the effectiveness of our method. The measured data include one wheeled vehicle and one tracked vehicle. There are three data collection scenarios, i.e., (a) vehicle first recedes from radar, then turns around and approaches radar; (b) vehicle recedes from radar and approaches radar; (c) vehicle circles around (for tracked vehicle only). In the data collection process, radar operates in a fixed position and all the measured data are collected under high SNR condition.

Two Doppler spectrum groups are shown in Figure 4. The wheeled vehicle has inconspicuous micro-Doppler components due to its rubbery wheel, and its bulk motion components dominate the Doppler domain. Unlike the wheeled vehicle, the tracked vehicle has a strong scattering structure, hence the clear micro-Doppler components are observed in the Doppler domain.

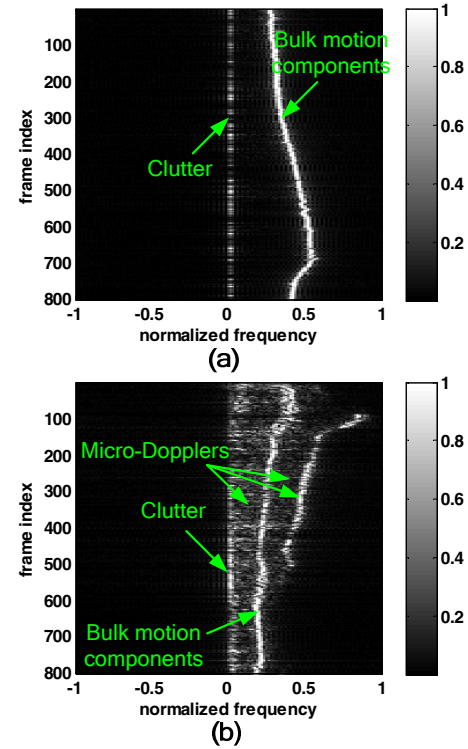


Figure 4. Doppler spectrum Groups from the measured data. (a) Wheeled vehicle. (b) Tracked vehicle.

The performances of noise reduction and super-resolution are shown in Figure 5. As shown in Figure 5b the super-resolution spectrum and noise reduction result are achieved with the sparse representation approach introduced in our work.

When the sparse representation is obtained, the effective features need to be extracted to train the selected classifier. Based on the observation that micro-Doppler frequencies induced by wheeled vehicle are inconspicuous and those induced by tracked vehicle are remarkable, we employ 4 features to describe the discriminative characteristic. These features are: (1) The spectrum entropy [4]. (2) The amplitude ratio of the bulk motion component to the maximum micro-

Doppler component. (3) The amplitude ratio of the bulk motion component to the average of the micro-Doppler components. (4) The energy ratio of the bulk motion component to the micro-Doppler components.

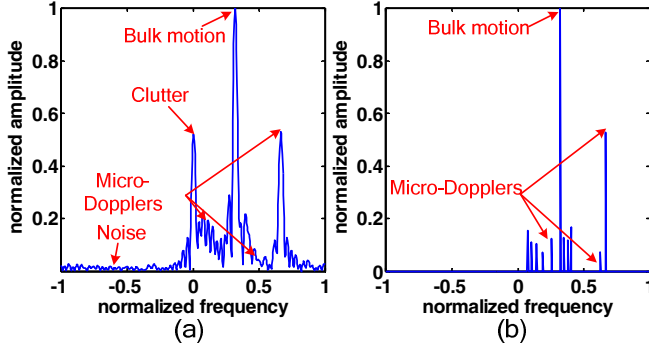


Figure 5. Performances of noise reduction and super-resolution. (a) Original Doppler spectrum. (b) super-resolution Doppler spectrum.

In the classification experiments, we utilize the measured data from scenario (a) as the training set, which are obtained under high SNR condition, and those from scenario (b) and (c) to get the test set under different SNR conditions, which are added the white noise with different noise power. The SNR here is measured for each slow time sample. We compare our method with a related method which is also proposed for radar target classification within short dwell time, i.e., the Principal Components Analysis (PCA) based method [3]. We also give the original spectrum based results, in which the aforementioned features are extracted from the original Doppler spectrum directly. For all the methods, support vector machine (SVM) [9] is employed as the classifier for fair comparison.

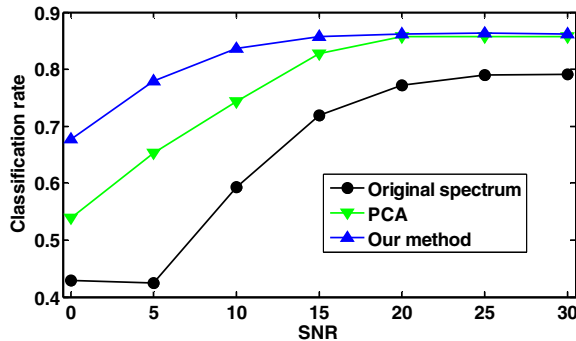


Figure 6. Classification performance of different methods versus SNR.

The classification performances versus SNR are shown in Figure 6. Since the influence of noise is relative nonsignificant under high SNR conditions ($\text{SNR} \geq 15\text{dB}$), the micro-Doppler information can be clearly extracted and good classification performance can be obtained for PCA based and our methods. It should be noticed that our method is superior to that of the original spectrum based method, which shows the super-resolution benefits the extraction of the micro-Doppler information. When SNR descends below 15dB, the noise influence is increasingly significant and micro-Doppler information is contaminated. Thus the performances of the related methods drop rapidly due to the lack of denoising

mechanism, whereas our method can remain relatively good under the low SNR conditions.

V. CONCLUSIONS

For moving vehicle classification within short dwell time, the discriminative information extraction and noise reduction should be considered together to obtain desired performance. In this paper, the sparse reconstruction of Doppler spectrum via l_1 -norm constrained optimization is utilized to obtain denoised super-resolution spectrum. Accordingly, the undesired influence of noise on classification performance is reduced, and more micro-Doppler information is extracted from the super-resolution spectrum. Thus good classification performance and noise robustness is achieved with the proposed classification scheme.

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